

MR BLS Trainer: A Physical Mixed Reality CPR+AED Rescue Simulator

Toshikazu Ohshima*
College of Image Arts and Sciences,
Ritsumeikan University, Kyoto, Japan
ohshimat@im.ritsumei.ac.jp

Saina Matsui
College of Image Arts and Sciences,
Ritsumeikan University, Kyoto, Japan
im0191xh@ed.ritsumei.ac.jp

Mizuki Yamane
College of Image Arts and Sciences,
Ritsumeikan University, Kyoto, Japan
im0194kh@ed.ritsumei.ac.jp

Yali Ling
College of Image Arts and Sciences,
Ritsumeikan University, Kyoto, Japan
im0196eh@ed.ritsumei.ac.jp

Katsuhito Muroi
College of Image Arts and Sciences,
Ritsumeikan University, Kyoto, Japan
im0192ep@ed.ritsumei.ac.jp

Chihiro Sakai
College of Image Arts and Sciences,
Ritsumeikan University, Kyoto, Japan
im0185ei@ed.ritsumei.ac.jp

ABSTRACT

The MR BLS Rescue Trainer is a training system that allows users to experience realistic simulation with physical feeling in lifesaving procedures, that is BLS (basic life support), including CPR (cardiopulmonary resuscitation) and AED (Automated External Defibrillator) by applying physical mixed reality incorporated with partially tangible components

CCS CONCEPTS

• **Human-centered computing**; • **Applied computing** → Education; Computer-assisted instruction; Life and medical sciences;

KEYWORDS

Mixed Reality, Basic Life Support, Cardiopulmonary Resuscitation, Automated External Defibrillator, Physical Mixed Reality

ACM Reference Format:

Toshikazu Ohshima, Saina Matsui, Mizuki Yamane, Yali Ling, Katsuhito Muroi, and Chihiro Sakai. 2023. MR BLS Trainer: A Physical Mixed Reality CPR+AED Rescue Simulator. In *SIGGRAPH Asia 2023 XR (SA XR '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610549.3614591>

1 INTRODUCTION

BLS training is typically taught by an instructor using a life-sized upper-body mannequin with no arms. Such constraints prevent us from taking enough opportunity to learn BLS. To make more opportunities of learning BLS, we developed the MR BLS Rescue Trainer. Similar systems have been proposed, one of which is a simple visual system that superimposes a virtual situation onto a real scene [Kwon et al., 2014]. And other system projects image of additional virtual arms and legs onto a physical mannequin [Johnson et al., 2018]. The advantage of our system is that by using mixed reality, we aim to provide trainees with realistic experience

*Corresponding author.

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SA XR '23, December 12–15, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0316-4/23/12.

<https://doi.org/10.1145/3610549.3614591>

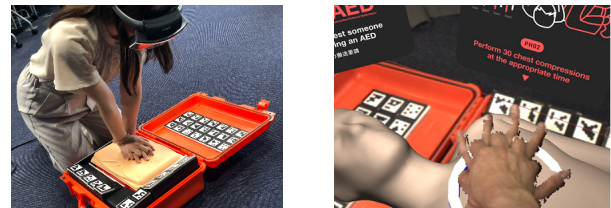


Figure 1: (a) Scene of the use (b) User's view

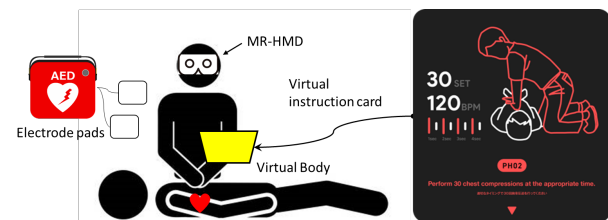


Figure 2: Image of the usage

and accurate knowledge on how to perform CPR and use an AED together. The experience includes physically heavy but precise work to press a chest rhythmically several minutes, with the minimum physical components, as shown in Figure 1. It will be possible to simulate various situations, such as emergency patients with different symptoms, ages, genders, and body sizes.

2 IMPLEMENTATION

Figure 2 shows an image of the usage of the system. The system consists of an HMD (Head Mounted Display), a human chest model for CPR training (Schooman2, Athena Industry), an AED simulator (Y283A, Nihon Kohden), and a PC. The human model and the AED simulator are integrated with a microcontroller and connected to the PC. The human model is placed on the floor. And then the user wears the HMD, kneels on the floor, and performs life-saving actions according to the detailed instructions displayed on the HMD. The displayed image on the HMD also can be seen by other audience via external monitors or projection on the wall.

Figure 3 depicts the system configuration. To show seamless mixed reality images, we use a video see-through type MR HMD

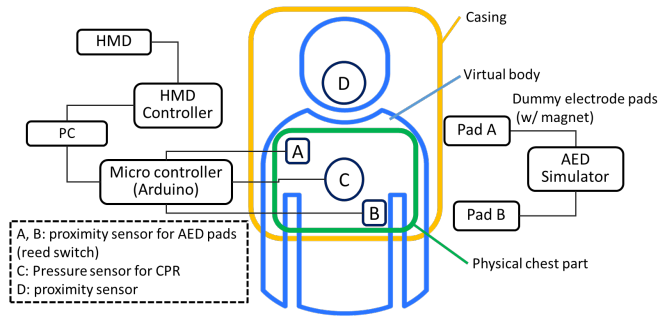


Figure 3: System configuration

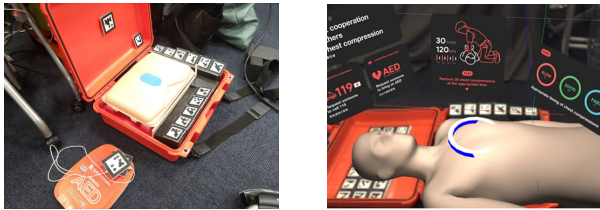


Figure 4: (a) Appearance of the system (b) User's view

(Canon MREAL HM-A1). Fiducial markers (Canon MREAL Marker) are also placed to track the position and rotation of the HMD.

The physical chest model for CPR training is obtained from an off-the-shelf consumer training kit and has a strong spring to provide realistic reaction force to the user. We modified the chest model with an additional pressure sensor that allows the system to assess whether the user is pushing sufficiently on the chest. In addition to that, magnetic reed switches are placed to detect electrode pads of AED are being attached on the chest model at correct positions.

The AED simulator we used is also an off-the-shelf consumer product. Originally the CPR kit and the AED kit were separate products, but we combined them with a microcontroller and sensors. They work together under a unified training scenario which is navigated by animated graphics. The chest model with the microcontroller and the fiducial markers are housed in a case with handle for portability and robustness. Figure 4 shows the appearance of the devices and user's view.

3 EXPERIENCE SCENARIO

The scenario comprises steps mainly as shown below (Figure 5):

- Ask people to call an ambulance and bring an AED
- Check the person for breathing
- CPR: Do chest compressions
- AED: Apply electroshock

The system provides people comprehensively basic lifesaving simulations by presenting the pace or timing of chest compressions, reproducing heartbeats, showing how to apply AED electrode pads, and visually presenting the necessary rescue activity guidance for each lifesaving phase as virtual instruction cards.

At each step, the virtual instruction is shown in front of the user. The virtual instructions are also provided to users as printed

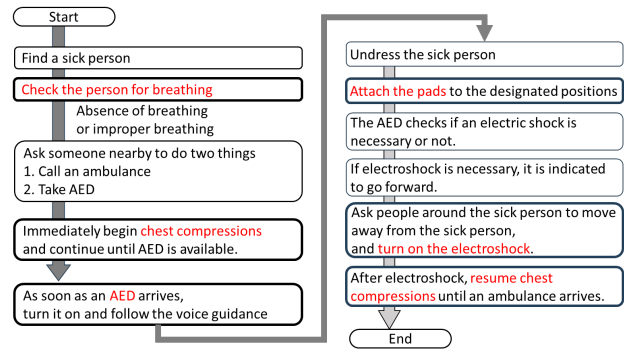


Figure 5: Experience scenario



Figure 6: (a) check breathing (b) CPR (c) AED use

cards to be able to review their learning after the mixed reality experience. The workflow is designed so that unless the rescue operation is performed correctly at each step according to virtual instructions, the rescue operation cannot proceed to the next step and the rescue operation will fail. Figure 6 shows scenes from the scenario; checking for breathing, compressing the chest, and ignition of electroshock by AED.

4 CONCLUSION

The MR BLS Trainer system allows users to easily perform lifesaving simulations without an instructor, and unlike conventional training experiences by using passive physical mannequin only, trainees will be able to learn the rescue procedure in various situations reproduced by the physical mixed reality.

Our future vision is to improve it so that in such a way that even children can experience it, by developing the system into a more compact and simpler, and with many simulation situations. And we would like to deploy the system into home and/or schools to make training activities more widespread in ordinary life.

ACKNOWLEDGMENTS

This work was supported by the Japan Society for the Promotion of Science KAKENHI, Grant Number JP21K12004.

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